

PRETORIA IRENE, South Africa

WMO: 682630

Lat: 25.9106S

Lon: 28.2106E

Elev: 1526

StdP: 84.29

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.8	4.0	-12.2	1.6	15.2	-10.3	1.9	14.7	9.1	13.4	8.3	14.1	3.3	180	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.8	31.3	16.1	30.1	16.1	29.0	16.3	20.2	26.0	19.6	25.2	19.2	24.6	4.3	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	16.2	21.9	18.1	15.7	21.6	17.6	15.2	21.4	65.6	26.0	63.6	25.3	61.8	24.5	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.3	6.4	DB	0.3	33.4	1.9	1.6	-1.0	34.6	-2.2	35.5	-3.2	36.4	-4.6	37.6	(4)
(5)			WB	-2.6	21.3	1.7	1.2	-3.8	22.2	-4.8	22.9	-5.8	23.6	-7.1	24.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.0	21.6	21.5	20.3	17.7	15.0	12.6	12.1	14.9	18.3	20.0	20.4	21.3	(6)	
	DBStd	4.27	2.17	2.00	2.27	2.43	2.44	2.26	2.48	3.18	3.50	3.24	2.92	2.44	(7)	
	HDD10.0	21	0	0	0	0	2	5	10	4	1	0	0	0	(8)	
	HDD18.3	714	3	2	8	40	103	171	194	112	43	21	13	4	(9)	
	CDD10.0	2926	359	323	320	230	158	83	74	156	250	310	313	351	(10)	
	CDD18.3	578	104	91	69	20	1	0	0	6	42	72	76	96	(11)	
	CDH23.3	4245	629	556	447	149	26	3	2	81	436	680	590	648	(12)	
(13)	CDH26.7	1026	151	124	92	16	1	0	0	5	103	200	158	176	(13)	
(14)	Wind	WSAvg	3.1	3.1	2.9	2.8	2.7	2.8	2.9	3.1	3.5	3.6	3.6	3.3	(14)	
(15) Precipitation	PrecAvg	691	125	104	96	39	16	7	2	4	15	60	96	127	(15)	
	PrecMax	1123	231	309	327	117	98	33	13	30	72	167	199	190	(16)	
	PrecMin	448	45	23	17	1	0	0	0	0	0	10	18	44	(17)	
	PrecStd	168	51	67	68	29	23	9	3	7	19	41	42	33	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.2	31.6	30.7	28.5	25.6	23.3	23.2	27.2	31.0	32.6	32.1	32.7	(19)	
		MCWB	17.3	17.2	16.3	16.0	12.7	10.1	9.2	12.4	13.3	14.5	15.5	17.8	(20)	
	2%	DB	30.0	29.6	29.0	26.8	24.0	21.6	21.6	25.5	29.3	30.6	30.1	30.5	(21)	
		MCWB	17.7	17.7	16.9	15.1	12.1	9.8	9.0	11.7	12.8	14.4	16.1	17.6	(22)	
	5%	DB	28.3	28.1	27.5	25.1	22.8	20.2	20.2	23.9	27.8	29.0	28.5	28.6	(23)	
		MCWB	18.0	17.9	16.7	15.0	11.9	9.6	8.9	11.1	12.6	14.5	16.1	17.7	(24)	
	10%	DB	26.9	26.8	25.9	23.5	21.4	18.9	18.8	22.2	26.1	27.3	26.9	26.9	(25)	
		MCWB	18.2	17.9	16.7	14.6	11.6	9.1	8.7	10.4	12.3	14.4	16.2	17.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.0	20.6	20.2	18.4	16.2	12.8	12.0	14.8	16.0	17.9	19.6	20.6	(27)	
		MCDB	27.3	26.7	25.4	23.9	21.4	18.9	18.6	22.7	24.8	25.0	26.2	26.5	(28)	
	2%	WB	20.2	19.9	19.3	17.6	14.4	11.7	11.0	13.6	15.1	17.1	18.8	19.7	(29)	
		MCDB	25.9	25.7	24.2	23.0	20.2	18.2	17.7	22.2	23.9	24.7	25.2	26.0	(30)	
	5%	WB	19.6	19.4	18.7	16.7	13.4	10.9	10.1	12.6	14.3	16.5	18.1	19.1	(31)	
		MCDB	25.0	24.8	23.5	21.8	19.4	17.4	17.1	20.7	23.3	24.1	24.5	25.2	(32)	
	10%	WB	19.1	18.9	18.0	15.9	12.6	10.2	9.4	11.6	13.6	15.9	17.5	18.6	(33)	
		MCDB	24.2	24.0	22.8	20.7	18.5	16.7	16.8	19.4	22.5	23.4	23.5	24.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	10.2	10.2	10.8	12.1	12.9	13.0	13.3	13.2	12.4	10.9	10.2	(35)	
		MCDBR	12.3	12.3	12.6	12.9	13.5	14.2	14.3	14.5	14.5	14.1	13.2	12.5	(36)	
	5% WB	MCWBR	4.3	4.2	4.4	4.7	5.5	6.2	6.1	6.1	5.4	4.8	4.3	4.1	(37)	
		MCWBR	9.9	9.8	9.8	10.1	11.2	12.4	12.2	12.7	12.9	12.1	10.9	10.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.389	0.391	0.380	0.333	0.297	0.279	0.281	0.335	0.375	0.408	0.370	0.377	(40)	
		taud	2.325	2.319	2.329	2.431	2.507	2.532	2.515	2.355	2.258	2.202	2.349	2.350	(41)	
	Ebn at Noon		954	936	916	917	913	912	923	901	906	910	966	968	(42)	
		Edh at Noon	137	135	127	105	88	81	86	111	134	150	133	135	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.63	6.29	5.57	4.77	4.38	3.97	4.31	5.07	6.03	6.47	6.73	6.78	(44)	
	RadStd		0.69	0.56	0.51	0.42	0.22	0.20	0.11	0.23	0.28	0.44	0.61	0.40	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	+0.66	+0.75	N/A	N/A	N/A	N/A	N/A	-181	+234	+87

Nomenclature: See separate page